SYED GHULAM MOEEN UD DIN SHAH BANOORI

AI Engineer | Agentic AI & RAG Systems | Computer Vision

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PROFILE

AI Engineer, Product Developer, and published researcher specializing in Agentic AI systems, Retrieval-Augmented Generation (RAG), computer vision, and applied machine learning. Combines a strong research foundation with peer-reviewed publications spanning generative AI for speech therapy and remote-sensing-based crop classification with hands-on experience turning research ideas into deployed, real-world AI products.

Skilled in taking projects from research and prototyping through to deployment, using tools such as YOLO, FaceNet, OpenCV, Transformers, Qdrant, Zep, and PostgreSQL. Experienced in designing local, cost-efficient AI architectures including memory-enabled agents and offline RAG pipelines that operate without dependence on paid LLM APIs, making them well suited to resource-constrained research and institutional settings.

Delivered AI-powered attendance, discipline-monitoring, and student-support systems for institutions including Fauji Foundation School and Edwardes College, alongside ongoing research in precision agriculture, IoT-based irrigation, and assistive AI communication tools for non-verbal children.

A trustworthy, detail-oriented collaborator who works effectively in research and engineering teams alike equally comfortable leading a team or independently driving a project from concept through to completion. Educated throughout in English medium institutions and proficient in English, with strong written and verbal communication demonstrated through published research papers and technical documentation.

EDUCATION

University of Engineering and Technology, Peshawar

August 2021 – July 2025

- Degree: Bachelor of Science in Computer Systems Engineering

WORK EXPERIENCE

AI Engineer *Thunder Energy & Intelligence | Jul 2025 – May 2026*

- Developing agentic AI systems and LLM-based solutions to enhance business workflows and decision-making.
- Implemented Zep memory management for the company's AI chatbot end-to-end in a single day; the integration worked seamlessly on first deployment, was presented internally by the company, and was praised by the client for its improved context retention.
- Reducing bot hallucinations and improving response reliability using Opik frameworks.
- Building and optimizing data pipelines to support telemetry analysis and Root Cause Analysis (RCA) of company operations.
- Automating queries and reports using PostgreSQL to improve efficiency and reduce manual workload.
- Collaborating on end-to-end AI solutions integrated with SaaS and PaaS platforms for scalable and reliable performance.
- Applying deep learning and machine learning models to analyze complex datasets and generate actionable insights.

Junior AI Engineer *Tadabur Tech | Jan 2025 – Jul 2025*

- Developing a face recognition model using Haar Cascade for face detection and Local Binary Patterns Histogram (LBPH) for recognition.
- Integrating the model with ESP-32 Cam for real-time image processing and authentication.
- Optimizing performance for efficient edge computing in IoT applications.

AI Developer – Intern *CODE INFORMATICS | Jul 2024 – Jan 2025*

- Conceptualized and contributed to an AI-driven solution targeting non-verbal children.
- Engineered and optimized computer vision models using OpenCV and YOLO, achieving high precision in object detection and face recognition tasks.

FEATURED PRODUCTS

AI Avatar Tutor & Offline RAG Assistant (2026 – Product)

product website

Developed an AI-powered avatar assistant capable of answering professor and student queries without relying on paid LLM APIs. Built a complete Retrieval-Augmented Generation (RAG) pipeline using local embeddings, Qdrant vector database, and local document storage.

- Integrated Qdrant vector database for semantic search and contextual retrieval.
- Enabled professors and doctors to upload books, PDFs, and academic documents for intelligent querying.
- Designed local document ingestion and storage pipelines for offline accessibility.
- Built a conversational AI avatar capable of explaining course material to students in the absence of instructors; implemented both text-to-speech (TTS) and speech-to-text (STT).
- Optimized for low-cost and institution-friendly deployment without recurring API costs.
- Implemented contextual memory and semantic retrieval for accurate educational assistance.

Algorithmic Cryptocurrency Trading Bot — Python, Pandas, CCXT, AsyncIO

- Designed and built a fully automated multi-timeframe trading bot analyzing 260,000+ candles of real BTC/USDT market data from Binance.
- Implemented technical analysis strategy using EMA-200 trend filters, Bollinger Band squeeze detection, and RSI momentum indicators across 5min/15min/1h timeframes.
- Built a professional risk management system with ATR-based dynamic stop-losses, trailing stops, position sizing, and portfolio correlation limits.
- Developed a realistic backtesting engine with next-bar execution, slippage modeling, and transaction fee simulation to eliminate look-ahead bias.
- Integrated a live market data pipeline via the Binance REST API with rate limiting and error handling.

AI Content Automation Platform — Python, FastAPI, React, FFmpeg, Whisper AI, SQLAlchemy

Product Website

- Built a self-hosted automation platform that transforms long-form videos into short-form social media content (YouTube Shorts, TikTok, Instagram Reels).
- Designed a modular, async architecture with plugin-based upload providers supporting multi-platform publishing.
- Integrated Faster-Whisper for speech-to-text, PySceneDetect for scene detection, and local LLMs (Ollama) for metadata generation.
- Implemented the full pipeline: video ingestion → transcription → AI clip detection → auto-editing → captions → thumbnail generation → upload.
- Built a REST API with FastAPI, background job processing, and real-time pipeline status tracking.

AI Camera-Based Attendance & Discipline Monitoring System (Schools & Offices)

Built and deployed a complete AI camera product for institutional attendance and discipline monitoring — combining a FaceNet-based desktop recognition application with real-world deployment at Fauji Foundation School, Chaklala Islamabad. Uses ESP32-CAM for live video capture, Haar Cascade for face detection, and FaceNet for recognition, with SQL-backed attendance logging.

- Automated attendance management using face recognition for students and staff.
- Uniform discipline monitoring using computer vision techniques.

- Real-time identity verification and reporting system, optimized for institutional-scale deployment.
- Desktop GUI with buttons for training, recognition, and dataset creation; efficient batching for face embedding storage and lookup.
- Daily auto-reset attendance logs with SQL integration for tracking.
- High accuracy on both known and unknown face classification.

AI-Assisted Communication Platform for Non-Verbal Autistic Children

Developing an AI-powered communication platform to help non-verbal autistic children express emotions, needs, and intentions through voice-enabled interactions.

- Implemented Speech-to-Text (STT) capabilities using Whisper to interpret and transcribe children's speech attempts and verbal inputs.
- Integrated Text-to-Speech (TTS) technologies, including ElevenLabs and Google TTS, to generate natural and engaging voice responses.
- Developed voice cloning functionality using ElevenLabs, enabling the system to provide encouragement, guidance, and positive reinforcement in the voice of a parent or caregiver, improving emotional connection and user engagement.
- Designed conversational AI workflows that allow children to communicate through voice, visual cues, and adaptive dialogue systems.
- Combined emotion detection, deep learning, and personalized interaction techniques to enhance communication between children, caregivers, therapists, and educators.
- Built adaptive feedback mechanisms that provide real-time assistance, speech practice, and interactive learning experiences.
- Focused on creating accessible, human-centered AI solutions that improve communication outcomes and social engagement for non-verbal children.

OTHER PROJECTS

AI Hostel Monitoring & Discipline System — Edwardes College Peshawar (Ongoing)

Currently developing an AI-powered hostel monitoring and discipline management platform for Edwardes College Peshawar.

- AI-assisted surveillance and behavioral monitoring.
- Hostelite discipline tracking and incident reporting.
- Intelligent analytics dashboard for administration.
- Integration of computer vision and AI-based monitoring workflows.

Agentic AI Memory Management

- Implemented long-term and short-term memory for AI systems using Zep, including session graphs and user context tracking.
- Leveraged Zep's dashboard to monitor and optimize memory utilization and system performance.

Chat-Bot Hallucination Mitigation & Evaluation

- Used Opik to reduce hallucinations in AI responses and improve reliability.
- Conducted prompt evaluation and answer evaluation using different ratios to measure and enhance response accuracy.

LLM Data Preprocessing Pipeline for Generative AI

- Developed an end-to-end data pipeline for training Large Language Models (LLMs).
- Performed data cleaning, tokenization (word-level, character-level, Byte Pair Encoding), vocabulary generation, and token ID mapping.
- Created input-target sequence pairs with positional encodings for effective sequence modeling.
- Implemented special-case tokens for unknown (out-of-vocabulary) words.

- Generated embeddings for use in transformer-based architectures for text generation tasks.
- Implemented causal attention.

Other Notable Projects

- **Deep Vertical Irrigation System using IoT and ML:** Developed a smart irrigation system utilizing IoT sensors and machine learning for efficient water usage.
- **Time Travel Bot:** Created a chatbot using advanced AI models and vector databases for conversational AI.
- **Image Captioning:** Developed a model to generate descriptive captions for images, enhancing accessibility.
- **Face Detection:** Designed a real-time face detection system using OpenCV and YOLO.
- **Emotion Detection:** Built an AI model for detecting emotions from images and real-time video streams.
- **Breast Cancer Classification:** Achieved 97.3% accuracy in detecting breast cancer using neural networks.
- **Chatbot Development:** Built a chatbot utilizing Mistral LLM integrated with Weaviate vector databases for retrieval-augmented generation (RAG).

RESEARCH WORK

Development of a Generative AI Application for Therapy of Speech-Impaired Patients

Published in the International Journal of Innovations in Science & Technology, Vol. 7 No. 2 (2025)

Designed a web-based application leveraging generative AI and voice recognition to provide remote speech therapy for individuals with speech impairments. The system delivers guided exercises, personalized feedback, and performance tracking to improve articulation, fluency, and pronunciation. Aims to expand access to affordable, AI-driven therapy in underserved regions through interactive and adaptive technology.

journal.50sea.com/index.php/IJIST/article/view/1383

Impact of Different Feature Engineering Techniques for Better Classification of Diverse Crops with Sentinel-2 Imagery

Published in the International Journal of Innovations in Science & Technology, Vol. 7 No. 3 (2025)

Conducted research on feature engineering approaches to improve crop classification using Sentinel-2 satellite imagery. Implemented and compared machine learning models such as Random Forest and Gradient Boosting, achieving high classification accuracy across multiple crop types. Focuses on enhancing agricultural analytics and precision farming using remote sensing and AI-based techniques.

journal.50sea.com/index.php/IJIST/article/view/1414

Smart Deep Irrigation System Using Machine Learning and IoT (Ongoing Research)

Developing a precision irrigation system that utilizes IoT sensors and machine learning algorithms to deliver optimal water levels based on real-time soil moisture, crop type, and weather data. The project includes a planned extension to vertical irrigation, targeting space-efficient urban farming and multi-level plantation setups by controlling water flow across different vertical layers. Aims to improve agricultural sustainability, reduce water consumption, and boost crop productivity through intelligent control systems.

SKILLS & INTERESTS

Skills:

- **Agentic AI & LLM Frameworks:** CrewAI, MCP, Zep, Opik, LangChain, Ollama
- **Retrieval-Augmented Generation (RAG):** Qdrant, Weaviate, Vector Databases, Semantic Search
- **AI & Deep Learning:** CNNs, YOLO, FaceNet, Transformers, OpenCV, Whisper AI
- **Machine Learning:** Predictive Modeling, Classification, Feature Engineering
- **Algorithmic Trading:** CCXT, Binance API, Backtesting, Technical Indicators, Risk Management
- **Programming Languages:** Python, C++, MATLAB
- **Databases & Backend:** PostgreSQL, SQL, pgAdmin, FastAPI
- **Deployment & Infrastructure:** Docker, Local AI Deployment, SaaS/PaaS Integration

- **Data Processing:** Tokenization, Embeddings, Data Pipelines, NLP Preprocessing, FFmpeg
- **Computer Vision:** Face Recognition, Emotion Detection, Surveillance Systems
- **Team & Leadership:** Team Leadership, Cross-Functional Collaboration, End-to-End Project Ownership
- **Languages:** English (Proficient educated in English-medium school and college), Urdu (Native)

Interests:

- AI, programming, writing novels, poetry, reading, and cricket.

CERTIFICATIONS

- Introduction to Model Context Protocol (MCP) — Anthropic (2026)
- MCP Mastery Certification — Fractal (2026)
- Docker Fundamentals — LearnKartS (2026)
- Multi AI Agent Systems with CrewAI — DeepLearning.AI (Completed July 10, 2025)
- Master RAG — Udemy (Paulo Dichone)
- Deep Learning Specialization — DeepLearning.AI
- Machine Learning Specialization — DeepLearning.AI
- Python (3-week course) — IBM via Coursera

MEDIA FEATURES

Recognized by leading media platforms for contributions to AI-driven hardware innovation:

Discover Pakistan Featured for building a smart real-time face recognition camera using AI and IoT
facebook.com/share/v/16Dma5HFFL

Voice of KP Interviewed for impactful work in tech and hardware innovation in Khyber Pakhtunkhwa
facebook.com/share/v/1DfWx9MEdH

Dialogue Pakistan Highlighted as a rising tech innovator for pioneering smart surveillance solutions
facebook.com/share/v/1HXK1cvzPa